FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
<u>FULL TEST – II</u>
PAPER –2
<u>TEST DATE: 04-01-2022</u>

Time Allotted: 3 Hours **Maximum Marks: 180**

<u>General Instructions:</u>

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

  Section – A ***(01 –06, 20 – 25, 39 – 44)*****:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

  Section – A ***(07 –10, 26 – 29, 45 – 48)*****:** This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

  Section – B ***(11 – 13, 30 – 32, 49 – 51)*****:** This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

  Section – C ***(14 – 19, 33 – 38, 52 – 57)*****:** This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

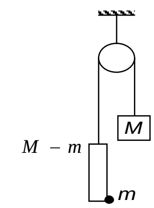
Physics

PART – I

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

1. A rod of mass '$M - m$' carries an insect of mass 'm' at its bottom end and its top end is connected with a string which passes over a smooth pulley and the other end of the string is connected to a counter mass M. Initially the insect is at rest. Choose the correct option(s).



(A) As insect starts moving up relative to rod, the acceleration of centre of mass of the system (insect + rod + counter mass) becomes non-zero

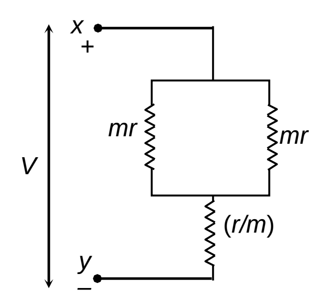
(B) As insect starts moving up relative to rod, tension in the string remains constant and is equal to Mg.

(C) As insect starts moving up relative to rod, the tension in the string becomes more than Mg.

(D) Acceleration of centre of mass of the system (insect + rod +counter mass) is zero when insect moves with constant velocity.

Ans. ACD

2. In the given circuit the value of m is varying. The correct statements about the circuit are



(A) The condition for maximum current flowing from x is m =2.

(B) The maximum current is $\frac{V}{\sqrt{2}r}$.

(C) The condition for maximum current flowing from x is $m = \sqrt{2}$.

(D) The maximum current is $\frac{3V}{2r}$.

Ans. BC

Sol. $i = \dfrac{2mV}{r\left(m^2+2\right)}$

$\dfrac{di}{dm} = 0 \Rightarrow m = \sqrt{2}$

$and\ i = \dfrac{V}{\sqrt{2}v}$

3. A triangle is made from thin insulating rods of different lengths; the linear charge density on each rod is uniform and the same for all three. At which point in the plane of the triangle at which the electric field strength is zero.

(A) Incentre of the triangle

(B) Centroid of the triangle

(C) Circumcenter of the triangle

(D) None of these

Ans. A

Sol. The electric field due to uniform line charge at a point is equal to the electric field due to a circular arc centered at that point which touches the line charge and limited by the ends of the line charge.

4. A uniformly charged insulating square sheet of side length d has charge Q. A point charge q is placed at a distance a/2 normally from the intersection of diagonals of the sheet. How much

(A) The force of electrostatic repulsion acts on the point charge due to the sheet is $\dfrac{Qq}{2\varepsilon_0 d^2}$.

(B) The force of electrostatic repulsion acts on the point charge due to the sheet is $\dfrac{Qq}{6\varepsilon_0 d^2}$.

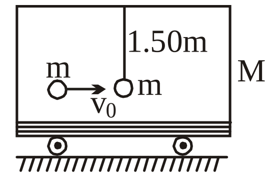
(C) The flux of electric field of the point charge that passes through the sheet is $\dfrac{Q}{6\varepsilon_0}$.

(D) The flux of electric field of the point charge that passes through the sheet is $\dfrac{Q}{12\varepsilon_0}$.

Ans. BC

Sol. Force on the sheet due to the point charge = charge density * electric flux passing through the sheet.

5. A ball of mass m = 1kg is hung vertically by a thread of length ℓ = 1.50 m. Upper end of the thread is attached to the ceiling of a trolley of mass M = 4kg. Initially, trolley is stationary and it is free to move along horizontal rails without friction. A shell of mass m = 1kg, moving horizontally with velocity $v_0 = 6\,ms^{-1}$, collides with the ball and gets stuck with it. As a result, thread starts to deflect towards right (g = 10 ms^{-2})



(A) The maximum deflection of the thread with the vertical is 37°

(B) The maximum deflection of the thread with the vertical is 45°

(C) The velocity of the trolley at that instant of maximum inclination of thread is 1 ms^{-1}

(D) The velocity of the trolley at that instant of maximum inclination of thread is 2 ms^{-1}

Ans. AC

Sol. When shell strikes the ball and gets stuck with it, combined body of mass 2m starts to move to the right. Let velocity of combined body (just after collision) be v_1.
According to law of conservation of momentum,
$(m + m)\,v_1 = mv_0$ or $v_1 = \frac{v_0}{2} = 3\ ms^{-1}$.
As soon as the combined body starts to move rightwards, thread becomes inclined to the vertical. Horizontal component of its tension retards the combined body while trolley accelerates rightwards due to the same component of tension.
Inclination of thread with the vertical continues to increase till velocities of both (combined body and trolley) become identical or combined body comes to rest relative to trolley. Let velocity at that instant of maximum inclination of thread be v. According to law of conservation of momentum.
$(2m + M)\,v = 2m.v_1$ or $v = 1\ ms^{-1}$
During collision of ball and shell, a part of energy is lost. But after that, there is no loss of energy. Hence, after collision, kinetic energy lost is used up in increasing gravitational potential energy of the combined body.
If maximum inclination of thread with the vertical be θ then according to law of conservation of energy,

$$\frac{1}{2}(2m)\,v_1^2 - \frac{1}{2}(2m + M)\,v^2 = 2mg\,(\ell - \ell\cos\theta)$$

$\therefore$ $\cos\theta = 0.8$ or $\theta = 37°$

6. If 1 kg of water freezes, then 334 kJ of energy are released. The energy released if 1 kg of super cooled water at $-10°C$ freezes is Q. Assume that during the freezing process the temperature remains constant, and the specific heat of water is $4.2\,kJ\,kg^{-1}K^{-1}$ and constant and specific heat of ice is $2.1\,kJ\,kg^{-1}K^{-1}$ which is also constant?

(A) The value of Q is 313 kJ

(B) The value of Q is 334 kJ

(C) The internal energy of ice obtained by this process is not equal to the internal energy of ice obtained by freezing water at 0^0 C and then cooled upto $-10°C$.

(D) The internal energy of ice obtained by this process is equal to the internal energy of ice obtained by freezing water at 0^0 C and then cooled upto $-10°C$.

Ans. AD

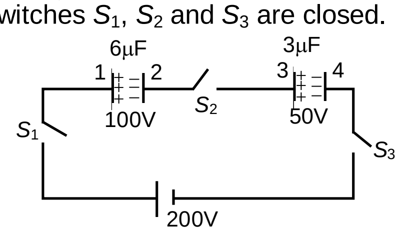
Sol. We could first carefully warm up the super cooled water from −10°C to 0°C, and then freeze the water at its normal freezing point. In the first part of this process, the heat that has to be supplied is 42 kJ, during the freezing stage the heat released is 334 kJ.
$-Q$+21 kJ = 42 kJ−334 kJ.
From this, Q = 313 kJ.

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 07 and 08

Two capacitors of capacity 6 μF and 3μF are charged to 100V and 50V separately and connected as shown in figure. Now all the three switches S_1, S_2 and S_3 are closed.



7. Charges on the 6μF capacitor in steady state will be

(A) 400 μC

(B) 700 μC

(C) 800 μC

(D) 250 μC

Ans. B

8. Suppose q_1, q_2 and q_3 be the magnitudes of charges flown from switches S_1, S_2 and S_3 after they are closed. Then

(A) $q_1 = q_3$ and $q_2 = 0$

(B) $q_1 = q_3 = \frac{q_2}{2}$

(C) $q_1 = q_3 = 2q_2$

(D) $q_1 = q_2 = q_3$

Ans. D

Sol. **(Q.7 – 8)**

$$\frac{600+q}{6}+\frac{150+q}{3}=200$$

$$100+\frac{q}{6}+50+\frac{q}{3}=200$$

$$\frac{q}{2}=50 \Rightarrow q=100\mu c$$

Charge on $6\mu F$ capacitor $=700\mu c$

From conservation of charge $q_1=q_2=q_3$

Paragraph for Question Nos. 09 and 10

In physics laboratory, Peter is trying to calculate the focal lengths of a convex lens. He measures the distance between a screen and a light source lined up on the optical bench to be 120 cm. When Peter shifts the convex lens along the axis of optical bench, sharp images of the source is obtained at two lens positions. He also measures the ratio of these two magnifications to be 1 : 9.

9. The focal length of the convex lens measured by Peter is

(A) 22.5 cm

(B) 30 cm

(C) 45 cm

(D) none of these

Ans. A

10. Which image, as seen by Peter is brighter?

(A) smaller

(B) bigger

(C) both are equally bright

(D) cannot be judged

Ans. B

Sol. **(Q.9 – 10)**

$$v+u=120............(1)$$

$$\frac{m_1}{m_2}=\frac{I_1}{I_2}=\frac{v^2}{u^2}=\frac{1}{9}\Rightarrow\frac{v}{u}=\frac{1}{3}.......(2)$$

So $v=30\,cm\, and\, u=90\,cm$ so $f=22.5\,cm$

As more light passes through lens, so bigger image is brighter

Section – B (Maximum Marks: 12)

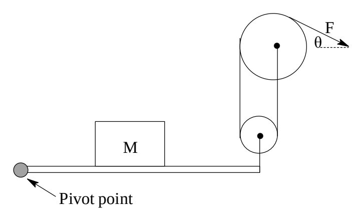
*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

11. Ann is sitting on the edge of a Disc that has a radius of 6 m and is rotating steadily. Bob is standing still on the ground at a point that is 12m from the centre of the carousel. At a particular instant, Bob observes Ann moving directly towards him with a speed of 1ms^{-1}. With what speed (in m/s) does Ann observe Bob to be moving at that same moment?

Ans. 2

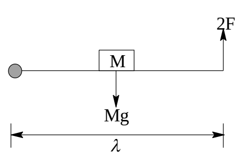
Sol. Consider the motions of the bodies both in the original frame of reference and in another frame that moves with velocity v_0 relative to the first one. Apply the Relative motion formulae. If Ann were sitting at the centre of the disc, she would see the whole world around her rotating with the same angular speed ω, but in the opposite direction. That means she would observe Bob standing 12 m away from the centre of the disc, but moving with a speed of 1/6 ×12 = 2 m s^{-1} in a direction perpendicular to the line joining him to the centre of the disc. Although Ann is not sitting at the centre of the disc, but at its edge, the same conclusion applies.

12. A pulley system is attached to a massless board as shown below. The board pivots only at the pivot point. A 10 kg mass M sits exactly in the middle of the board. If the angle θ is 60°. The force F (in N) necessary to hold the board in the position shown is $5\times x$, find x?



Ans. 5

Sol.



For equilibrium

$$\left|Mg\frac{\lambda}{2}\right| = |2F|1$$

$$F = \frac{mg}{4} = \frac{100}{4} = 25\,N$$

13. The peak emission from a black body at a certain temperature occurs at a wavelength of 6000 Å. On increasing its temperature, the total radiation emitted is increased 16 times. These radiations are allowed to fall on a metal surface. Photoelectrons emitted by the peak radiation at higher temperature can be bought to rest by applying a potential equivalent to the excitation potential corresponding to the transition for the level n = 4 to n = 2 in the Bohr's hydrogen atom. The work function of the metal is given by $\frac{\alpha}{100}eV$ where α is the numerical constant. The value of α is found $53\times x$, find x? [Take: he = 12420 eV-Å]

Ans. 3

Section – C (Maximum Marks: 12)

This section contains ***THREE (03)*** *question stems. There are* ***TWO (02)*** *questions corresponding to each question stem. The answer to each question is a* ***NUMERICAL VALUE.*** *If the numerical value has more than two decimal places, truncate/round-off the value to* ***TWO*** *decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

A quantity, *n* mol say, of helium gas undergoes a thermodynamic process (neither isothermal nor isobaric) in which the molar heat capacity *C* of the gas can be described as a function of the absolute temperature by $C = 3RT/(4T_0)$, where *R* is the gas constant and T_0 is the initial temperature of the helium gas.

14 The temperature of the gas, when is reaches its minimal volume is nT_0, find the value of n.

Ans. 2.00

15. The work done on the system up until the point at which the helium gas reaches its minimal volume is $\left(\frac{3}{n}\right)nRT_0$, find the value of n.

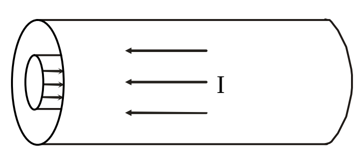
Ans. 8.00

Sol. **(Q.14 – 15)**
When the helium gas is close to its minimal volume, to first order there is no change in the volume for a finite change in the temperature. This means that, in this region, the process is approximately isochoric, and so the molar heat capacity of the monatomic helium gas must be C_V

Question Stem for Question Nos. 16 and 17

Question Stem

A constant current I flows through a cable consisting of two thin co-axial metallic cylinders of radii R and 2R, as shown in figure



16. If the inductance per unit length is $\frac{\mu_0}{2\pi}\log_e n$. Then find the value of n.

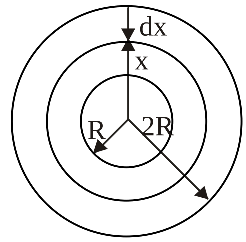
Ans. 2.00

17. If the energy stored in it per unit length and $\frac{\mu_0 I^2}{4\pi}\log_e n$. Then find the value of n.

Ans. 2.00

Sol. **(Q.16 – 17)**
Due to flow of current through the cable, shown in fig. magnetic field is established in the space between cylinders. Energy is stored in this space due to magnetic field.

Since, energy stored per unit volume in magnetic field of induction B is equal to $B^2 / 2\mu_0$, therefore, magnetic induction in the space must be known. But magnetic field in the space is not uniform. Hence, consider a thin cylindrical coaxial shell in the space. Let radius of the shell be x and radial thickness dx as shown in fig.
According to Ampere's circuital law, magnetic induction B is given by $B\, 2\pi x = \mu_0 I$



or $B = \dfrac{\mu_0 \pi I}{2\pi x}$

$\therefore$ Energy stored per unit length of the shell considered is

$$dU = \frac{B^2}{2\mu_0}(2\pi x\, dx) = \frac{\mu_0 I^2}{4\pi x} dx$$

$\therefore$Energy stored per unit length of the cable,

$$U = \int dU = \int_{x+r}^{x=2R} \frac{\mu_0 I^2}{4\pi x} dx$$

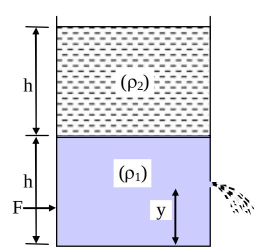
$$= \frac{\mu_0 I^2}{4\pi} \log_e 2$$

Since, energy stored in magnetic field is $U = \dfrac{1}{2} LI^2$ where L is self inductance, therefore, inductance per unit length of the cable is $L = \dfrac{2U}{I^2} = \dfrac{\mu_0}{2\pi} \log_e 2$

Question Stem for Question Nos. 18 and 19

Question Stem

A cylindrical tank is filled with two liquids of density ρ_1 = 900 kg m^{-3} & ρ_2 = 600 kgm^{-3}, to a height h = 60 cm each as shown in figure. A small hole having area a = 4.9 cm^2 is made in right vertical wall at a height y = 20 cm from the bottom. (g = 10 m/s^2)



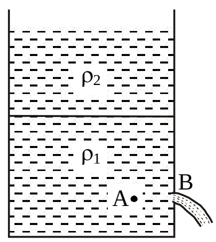
18. The velocity of efflux is (in m/s)

Ans. 4.00

19. The horizontal force F to keep the cylinder in static equilibrium, if it is placed on a smooth horizontal plane is (in N)

Ans. 7.00

Sol. **(Q.18 – 19)**
Since area a of hole is very small in comparison to base area A of the cylinder, therefore, velocity of liquid inside the cylinder is negligible. Let velocity of efflux be v and atmospheric pressure P_0. Consider two points A (inside the cylinder) and B (just outside the hole) in the same horizontal line as shown in figure.



Pressure at A, $P_A = P_0 + h\,\rho_2 g + (h - y)\rho_1 g$
Pressure at B, $P_B = P_0$
According to Bernoulli's theorem,
pressure energy at A
= pressure energy at B + kinetic energy at B

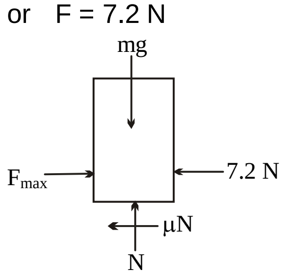
$\therefore \quad P_A = P_B + \frac{1}{2}\rho_1 v^2$

$\therefore \quad v = 4 ms^{-1}$ **Ans. (i)**

When cylinder is on smooth horizontal plane, force F required to keep cylinder stationary equals horizontal thrust exerted by water jet. But that is equal to mass flowing per second × change in velocity of this mass.

$\therefore \quad F = (av\,\rho)(v - 0) = a\rho v^2$

or $F = 7.2$ N **Ans.(ii)**



Total mass of the liquid in the cylinder is
$m = Ah\rho_1 + Ah\rho_2 = 450$ kg

Limiting friction = $\mu mg = 45$ N

$\therefore$ F < Limiting friction, therefore, minimum force required is zero. **Ans. (iii)**

Chemistry

PART – II

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

20. The vapour pressure of pure liquids A and B are 450 & 700 mm Hg respectively at 350K. If the total vapour pressure of solution (A+B) is 600 mm Hg, then which of the following is/are correct?

(A) Mole fraction of A in liquid mixture is 0.4

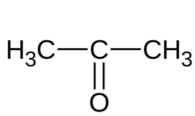
(B) Mole fraction of A in liquid mixture is 0.6

(C) Mole fraction of A in vapour phase is 0.3

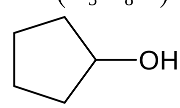
(D) Mole fraction of A in vapour phase is 0.35

Ans. AC

21. An optically active alcohol $A\left(C_8H_{16}O\right)$ on oxidation gives B. A on acidic heating gives $C\left(C_8H_{14}\right)$ as major product. C on ozonolysis produces $D\left(C_5H_8O\right)$ and



. D on reduction with LiAlH_4 gave



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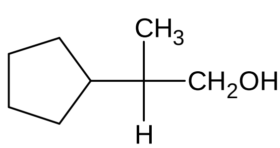
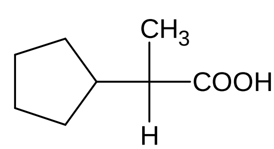
Identify correct answer:

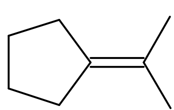
(A) A is



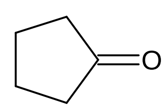
(B) B is



(C) C is

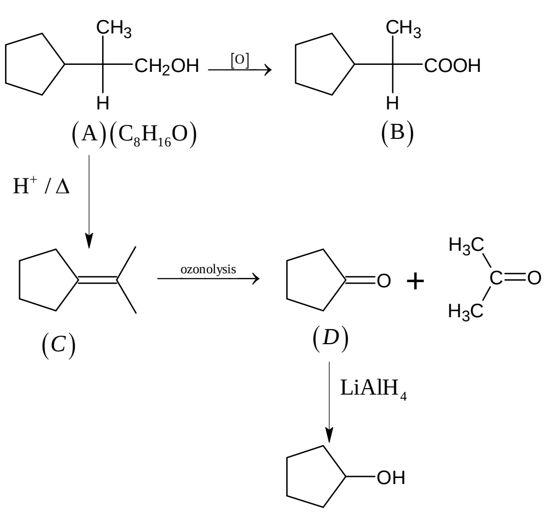


(D) D is

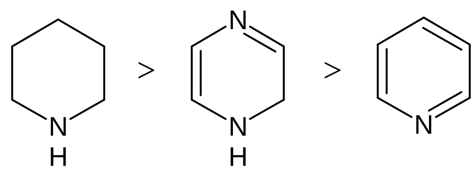


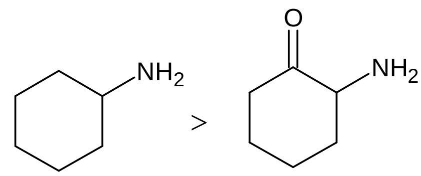
Ans. ABCD

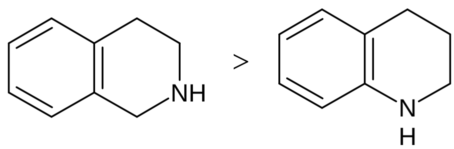
Sol.

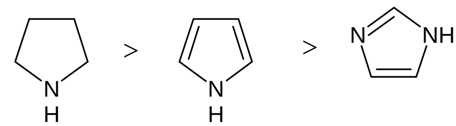


22. Which of the following order for basic strength is/are correct?

(A) 

(B) 

(C) 

(D) 

Ans. ABC

Sol. A lone pair of N-atom participating in resonance will be less basic

23. Which of the following is/are correct :

(A) Carbon-carbon bond length in CaC_2 will be more than in CH_2CCH_2

(B) O–O bond length in KO_2 will be more than in Na_2O_2 .

(C) O–O bond length in $O_2[PtF_6]$ will be less than that in KO_2

(D) N–O bond length in NO gaseous molecule will be smaller than in $NOCl$ gaseous molecule.

Ans. CD

24. Anode rays

(A) contain positively charged particles

(B) have constant e/m ratio irrespective of the type of gas used

(C) may contain He^+ ions

(D) are always beams of protons

Ans. AC

25. Which is correct for a gas described by van der Waal's equation:

(A) behaves similar to an ideal gas in the limit of large molar volumes

(B) behaves similar to an ideal gas in the limit of large pressures

(C) is characterized by van der Waal's coefficients that are dependent on the identity of the gas but are independent of the temperature.

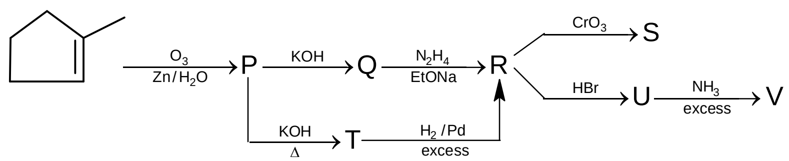
(D) has the pressure that is lower than the pressure exerted by the same gas behaving ideally

Ans. ACD

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 26 and 27



26. Which of the following statement is correct about S?

(A) It will not give yellow/orange or red colour with 2, 4-DNP

(B) It will give red colour with Cerric Ammonia Nitrate

(C) It will not give Fehling test

(D) None of these

Ans. C

27. Which of the following statement is correct about V?

(A) It will not give mustard oil reaction

(B) It will not give carbylamines test or isocyanide test

(C) It is less basic than aniline

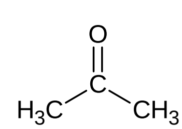
(D) It will give base soluble product with Hinsburg reagent

Ans. D

Paragraph for Question Nos. 28 and 29

An acyclic hydrocarbon P, having molecular formula C_6H_{10}, give acetone as the only organic product through the following sequence of reactions, in which Q is an intermediate organic compound.

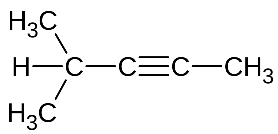
$$\underset{(C_6H_{10})}{P}\ \xrightarrow[\substack{(ii)\ NaBH_4/\text{ethanol} \\ (iii)\ \text{dil. acid}}]{(i)\ \text{dil.}H_2SO_4/HgSO_4}\ Q\ \xrightarrow[\substack{(ii)\ O_3 \\ (iii)\ Zn/H_2O}]{\substack{(i)\ \text{conc.}H_2SO_4 \\ (\text{Catalytic amount}) \\ (-H_2O)}}\ 2$$

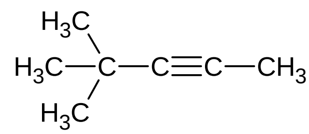


28. The structure of compounds P is:

(A) $\mathrm{CH_3}CH_2CH_2CH_2-C\equiv C-H$

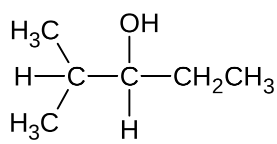
(B) $H_3CH_2C\equiv C-CH_2CH_3$

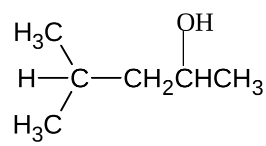
(C) 

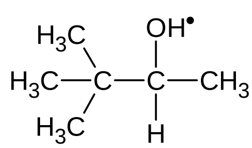
(D) 

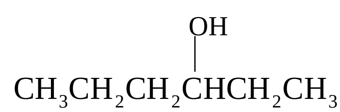
Ans. D

29. The structure of the compound Q is:

(A) 

(B) 

(C) 

(D) 

Ans. B

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

30. The number of electrons for Zn^{2+} cation that have the value of azimuthal quantum number $= 0$ is:

Ans. 6

Sol. Azimuthal quantum number is zero for s-subshell

31. Co forms dinuclear complex with a sigma bond within two Co atoms. Consider that metal carbonyls follows EAN rule. The complex can be written as $\mathrm{CO_2}\left(CO\right)_x$. Find the value of x ?

Ans. 8

Sol. $\left(CO\right)_{x/2}Co-Co\left(CO\right)_{x/2}$

$$EAN\left(=36\right)=27+1+2\times\frac{x}{2}$$

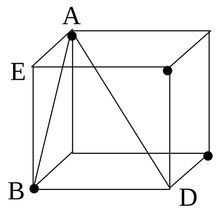
$$\Rightarrow 28+x=36$$

$$x=8$$

32. If four atoms of same radius are placed at the alternate corner of a cube touching each other, than the length of body diagonal of the cube is equal to $\sqrt{x}\times R$, where R is the radius of atom. Find the value of x?

Ans. 6

Sol.



Face diagonal of cube $(\mathrm{AB}) = 2\mathrm{R}$

Edge length of cube $(\mathrm{BE}) = \frac{2\mathrm{R}}{\sqrt{2}}$

Body diagonal of cube $(\mathrm{AD}) = \sqrt{3} \times \frac{2\mathrm{R}}{\sqrt{2}}$

$= \sqrt{6} \times R$

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem

An aluminium block of 270gm having total heat capacity 490 J/K. at 127°C is placed in a lake at 27°C. $\Delta \mathrm{S}_{\mathrm{universe}}$ is given by $x\,\mathrm{J/K}$. When the same block at 27°C is dropped in the lake (27°C) from a height of 200m. $\Delta \mathrm{S}_{\mathrm{universe}}$ is given by $\mathrm{y\,J/K}$. [$\ln(4/3) = 0.29, g = 10.0\mathrm{m/sec}^2$]

33. The value of x is_____.

Ans. 21.23

Sol. $\Delta \mathrm{S}_{\mathrm{lake}} = \frac{490 \times (400 - 300)}{300} = 163.33 \quad \Delta S_{black} = 490 \times \ln(300/400) = -142.1$

$x = \Delta S_{univ} = 163.33 - 142.1 = 21.23 J/K$

34. The value of y is_____.

Ans. 1.80

Sol. $\Delta S_{black} = 0; \quad \Delta S_{lake} = \frac{mgh}{300} = \frac{0.270 \times 10 \times 200}{300} = 1.8\,\mathrm{J/K}$

Question Stem for Question Nos. 35 and 36

Question Stem

A mixture of $NaCl$ & $NaBr$ weighing 4.0 gm was dissolved and treated with excess of $AgNO_3$ to precipitate all of the chlorides & bromides as AgCl and $AgBr$. The washed ppt was treated with KCN to dissolve the precipitates. The resulting solution was electrolysed, the weight of Ag deposited is 6.0 gm.

$$K\left[Ag(CN)_2\right] + KOH \xrightarrow{\text{electrolysis}} Ag + KCN + O_2 + H_2O$$

$$\left(\text{M.W. of NaCl, NaBr and Ag are } 58.5, 103, 108\,\text{g/mol}\right)$$

35. Wight percent of $NaCl$ in the original mixture is____.

Ans. 56.60

36. The volume $\left(\text{at STP, in ltr}\right)$ of O_2 produced during electrolysis is_____.

Ans. 0.31

Sol. (Q.35 – 36)

x mass of $NaCl$

$$\frac{x}{58.5} + \frac{4-x}{103} = \frac{6}{108} \Rightarrow x = 2.264$$

Moles of O_2 produced $= \frac{1}{4} \times \text{moles of Ag}$

Question Stem for Question Nos. 37 and 38

Question Stem

At 298 K, the limiting molar conductivity of a weak monobasic acid is $4\times10^2\ S\,cm^2 mol^{-1}$. At 298 K, for an aqueous solution of the acid the degree of dissociation is α and the molar conductivity is $y\times10^2\ S\,cm^2 mol^{-1}$. At 298 K, upon 20 times dilution with water, the molar conductivity of the solution becomes $3y\times10^2\ S\,cm^2 mol^{-1}$.

37. The value of α is____.

Ans. 0.22

Sol.

$$\alpha_1 = \frac{\lambda_m^c}{\lambda_m^0} = \frac{y\times10^2}{4\times10^2} = \frac{y}{4} = \alpha$$

$$\alpha_2 = \frac{\lambda_m^c}{\lambda_m^0} = \frac{3y\times10^2}{4\times10^2} = \frac{3y}{4} = 3\alpha$$

$$K_a = \frac{C_1\alpha_1^{\,2}}{1-\alpha_1} = \frac{C_2\alpha_2^{\,2}}{1-\alpha_2}$$

$$\frac{C\alpha^2}{1-\alpha} = \frac{(C/20)(3\alpha)^2}{1-3\alpha}$$
$$\alpha = 0.2156$$
$$\alpha = 0.22$$

38. The value of y is______.

Ans. 0.88

Sol. $\alpha = \frac{y}{4};\ y = 4\alpha = 4\times 0.22 = 0.88$

Mathematics

PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

39. The volume of a right triangular prism $ABCA_1B_1C_1$ is equal to 3. If the position vectors of the vertices of the base ABC are A(1, 0, 1) ; B(2,0, 0) and C(0, 1, 0) the position vectors of the vertex A_1 can be:

(A) (2, 2, 2)

(B) (0, 2, 0)

(C) (0, - 2, 2)

(D) (0, - 2, 0)

Ans. AD

Sol. knowing the volume of the prism we find its altitude H = (A A_1) = and designating the vertex A_1 (x_1, y_1, z_1) relate the co-ordinates of the vector $\vec{AA_1}$ = (x - 1, y, z - 1) and its length. We get the other equation from the condition $\vec{AA_1}$ perpendicular to $\vec{AC}$]

40. If the equation $|z|(z+1)^8 = z^8|z+1|$ where z $\in$ C and $z(z+1) \neq 0$ has distinct roots z_1, z_2, z_3, ..., zn (where n $\in$ N) then which of the following is/are true?

(A) $z_1, z_2, z_3, ..., z_n$ are concyclic points

(B) $z_1, z_2, z_3, ..., z_n$ are collinear points

(C) $\sum_{r=1}^{n} \text{Re}(z_r) = \frac{-7}{2}$

(D) $\sum_{r=1}^{n} \text{Im}(z_r) = 0$

Ans. BCD

41. If in a ΔABC, a, b, c are in A.P. and P_1, P_2, P_3 are the altitudes from the vertices A, B and C respectively, then

(A) P_1, P_2, P_3 are in A.P.

(B) P_1, P_2, P_3 are in H.P.

(C) $P_1 + P_2 + P_3 \leq \frac{3R}{\Delta}$

(D) $\frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3} \leq \frac{3R}{\Delta}$

Ans. BD

42. Suppose a_1, a_2, a_3 are in A.P. and b_1, b_2, b_3 are in H.P. and let $\Delta = \begin{vmatrix} a_1-b_1 & a_1-b_2 & a_1-b_3 \\ a_2-b_1 & a_2-b_2 & a_2-b_3 \\ a_3-b_1 & a_3-b_2 & a_3-b_3 \end{vmatrix}$, then

(A) Δ is independent of a_1, a_2, a_3, b_1, b_2, b_3

(B) $a_1 - \Delta$, $a_2 - 2\Delta$, $a_3 - 3\Delta$ are in A.P.

(C) $b_1 + \Delta$, $b_2 + \Delta^2$, $b_3 + \Delta$ are in H.P

(D) none of these

Ans. ABC

43. Let A, B, C be n×n real matrices and are pair wise commutative and ABC = O_n and if $\lambda = \det(A^3 + B^3 + C^3).\det(A + B + C)$ then

(A) $\lambda > 0$

(B) $\lambda < 0$

(C) $\lambda = 0$

(D) $\lambda \in (-\infty, \infty) - \{0\}$

Ans. AC

Sol. $A^3 + B^3 + C^3 = A^3 + B^3 + C^3 - 3ABC$

$= (A+B+C)(A^2+B^2+C^2-AB-BC-CA)$

$= (A+B+C)(A+WB+W^2C)(A+W^2B+WC)$ [W = cube root of unity]

$= (A+B+C)(A+WB+W^2C)\overline{(A+WB+W^2C)}$

thus $\det(A^3 + B^3 + C^3).\det(A + B + C)$

$\det(A + B + C)^2.\det(A + WB + W^2C)\det\overline{(A+WB+W^2C)}$

$=(\det(A+B+C))^2.\left|\det(A+WB+W^2C)\right|^2 \geq 0$.

44. Let f: A $\to$ A, where A is a set of non-negative integers and satisfying the condition.
(i) $x - f(x) = 19[x/19] - 90[f(x)/90]$, $\forall x \in A$
(ii) $1900 < f(1900) < 2000$
then f(1900) equals, where [ ] indenter greatest integer function

(A) 1994

(B) 1900

(C) 1904

(D) 1890

Ans. AC

Sol. Put x = 1990 in(i) we get $\left[\frac{f(1990)}{90}\right]$ = f(1990) – 14 ……….(0)(iii)

Using relation(ii) $\frac{1990}{90} < \frac{f(1990)}{90} < \frac{2000}{90}$

$\Rightarrow \left[\frac{f(1990)}{90}\right]$ = 21 or 22

of $\left[\frac{f(1990)}{90}\right]$ = 21 $\Rightarrow$ f(1990) = 1904 Using (iii)

of $\left[\frac{f(1990)}{90}\right]$ = 22 $\Rightarrow$ f(1990) = 1994 using (iii)

Section – A (Maximum Marks: 12)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

Paragraph for Question Nos. 45 and 46

Let t be a real number satisfying $2t^3 - 9t^2 + 30 - \lambda = 0$ where $t = x + \frac{1}{x}$ and $\lambda \in R$ then

45. If the cubic equation has three real and distinct solution for x then λ

(A) Is greater than 9

(B) Is greater than 11

(C) Is less than 8

(D) Is not equal to 10

Ans. A

Sol. For $\lambda = 10 \Rightarrow t_1 = 2, t_2 > 2$ and if $x + 1/x = 2 \Rightarrow x = 1$ and $x + 1/x = t_2$ so there will be two distinct values of x

46. If the cubic equation has exactly two real and distinct roots of x then exhaustive set of values of λ is

(A) $\lambda \in (-\infty, 3) \cup (30, \infty)$

(B) $\lambda \in (-\infty, -22) \cup (10, \infty) \cup \{3\}$

(C) $\lambda \in \{3, 30\}$

(D) None of these

Ans. B

Sol. Since domain of $f(t) = 2t^3 - 9t^2 + 30$, $t = x + 1/x$, $|t| \geq 2$ $f(-2) = -22$, $f(2) = 10$, critical points at $t = 0 \,\&\, 3$; $f(3) = 3$

Paragraph for Question Nos. 47 and 48

Let P, Q are two points on the curve y = $\log_{1/2}(x - 0.5) + \log_2 \sqrt{4x^2 - 4x + 1}$ and P is also on the circle $x^2 + y^2 = 10$, Q lies inside the given circle such that its abscissa is an integer.

47. The co–ordinates of P are given by

(A) (1, 2)

(B) (2, 4)

(C) (3, 1)

(D) (3, 5)

Ans. C

Sol. $y = \log_{1/2}\left(x - \frac{1}{2}\right) + \log_2 \sqrt{(2x-1)^2}$ But $x > \frac{1}{2}$

$= \log_{1/2}\left(x - \frac{1}{2}\right) + \log_2 (2x - 1)$

y = 1

P ≡ (3, 1).

48. $\overrightarrow{OP} \cdot \overrightarrow{OQ}$, 'O' being the origin is

(A) 4 or 7

(B) 4 or 2

(C) 2 or 3

(D) 7 or 8

Ans. A

Sol. $\overrightarrow{OP} = 3\hat{i} + \hat{j}$

Q ≡ (1, 1) or (2, 1)

$\overrightarrow{OQ} = \hat{i} + \hat{j}$ and $2\hat{i} + \hat{j}$

$\overrightarrow{OP} \cdot \overrightarrow{OQ} = 3 + 1 = 4$ and 6 + 1 = 7

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

49. Consider on equation with x as variable $7\sin 3x - 2\sin 9x = \sec^2\theta + 4\operatorname{cosec}^2\theta$ then the value of $\frac{15}{2\pi}$ [minimum positive root – maximum negative root] is:

Ans. 5

50. If the area enclosed by $g(x), x = -3, x = 5$ and x-axis where $g(x)$ is the inverse of $f(x) = x^3 + 3x + 1$ is A, then $[A]$ is ($[.]$ is G. I. F)

Ans. 4

51. Let $f(x) = \cos 2x.\cos 4x.\cos 6x.\cos 8x.\cos 10x$ then $\lim\limits_{x\to 0}\frac{1-(f(x))^3}{55\sin^2 x}$ equals

Ans. 6

Sol. $\lim\limits_{x\to 0}\frac{\ln\cos 2x}{x^2} = -\frac{2^2}{2}, \lim\limits_{x\to 0}\frac{\ln\cos 4x}{x^2} = -\frac{4^2}{2}$

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Let $f(x) = x^{2010} + x^{1010} - x^{510} + x^{210} + x^2$. If $f(x)$ is divided by $x^2(x^2-1)$, then we get remainder as $g(x) = ax^2 + bx^2 + cx + d$.

52. $|a+b+c+d|$ is equal to____.

Ans. 3.00

53. If roots of $g(x) = 0$ lies between the roots of the equation $x^2 - 2(a+1)x + a(a-1) = 0$ then the number of integral values of a will be_____.

Ans. 0.00

Question Stem for Question Nos. 54 and 55

Question Stem

Let $f(x)=x+\frac{2}{1.3}x^3+\frac{2.4}{1.35}x^5+\frac{2.4.6}{1.3.5.7}x^7+....\forall x\in(0,1)$

54. If the value of $f\left(\frac{1}{2}\right)$ is $\frac{\pi}{a\sqrt{b}}$ (where $a,\ b\in\mathbb{R}$) then $|a+b|$ is______.

Ans. 6.00

55. Let $g(x)=\sqrt{1-x^2}\,.f(x)$ for all $x\in(0,1)$. If the range of $g(x)$ is $\left(0,\frac{\pi}{c}\right)$ (where $c\in\mathbb{R}$) then c is_____.

Ans. 2.00

Question Stem for Question Nos. 56 and 57

Question Stem

Let z_1 and z_2 are two complex numbers such that $|z_1|=a$ and $|z_2|=b$ satisfying the equation $3z_1^{\ 2}-2z_1z_2+2z_1^{\ 2}=0$ and also $\mathrm{Re}\left(\frac{z_1-2}{z_1+2}\right)=0$. Also P,Q and O are points in complex plane corresponding to z_1, z_2 and origin respectively.

56. Eccentricity of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,$ is_____.
[Answer should be rounded off to two decimal planes]

Ans. 0.58

57. Area of triangle OPQ is______.
[Answer should be rounded off to two decimal planes]

Ans. 2.24